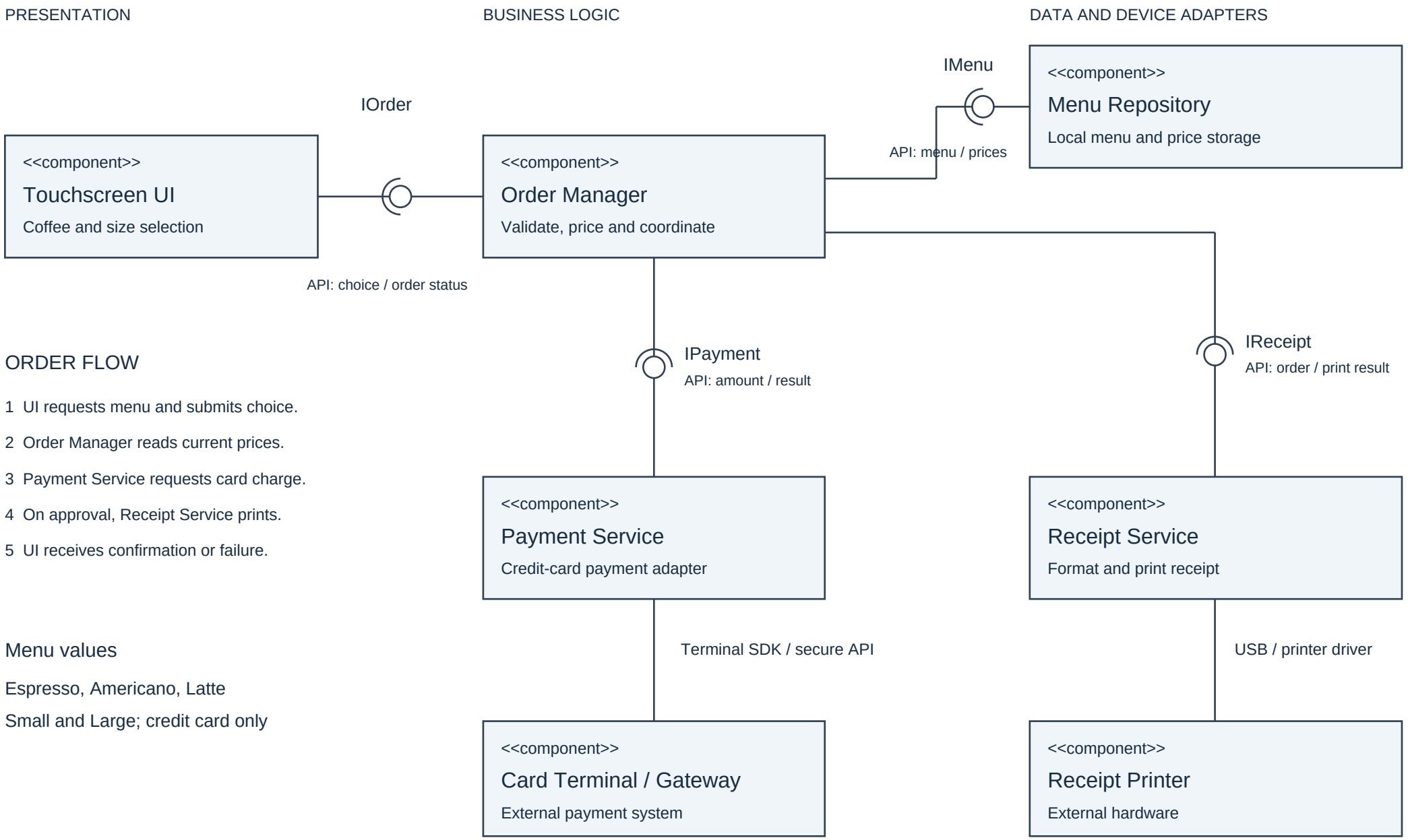
Lab 3 Component Modelling and Architectural Pattern Selection

Self-Service Coffee Kiosk System | PES1UG24AM234 | Layered architecture



Notation: circle = provided interface; semicircle = required interface; solid line = service connection.

The provider is at the right or bottom of each assembly connector. Calls return results to the requesting component.

Architecture Selection and Justification

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Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System. The touchscreen forms the presentation layer; the Order Manager coordinates business rules; the menu repository and payment and receipt adapters encapsulate storage and device integration. These are logical modules deployed together on the kiosk, with external payment and printer dependencies.

Architectural Style Analysis

Layered: Clear responsibilities, simple local deployment and low communication overhead suit a single kiosk with a small fixed menu. The tradeoff is a shared application failure boundary and limited independent scaling.
Microservices: Independent deployment and scaling could benefit a large kiosk fleet, but service discovery, network failures and distributed order/payment consistency add unnecessary complexity here.
Client-Server: A central server can simplify menu updates across several kiosks, but server or network failure can block ordering. It is less suitable for this standalone scenario. Layering and client-server deployment can coexist; layering is the primary design choice here.

Two Scenario Specific Reasons

1. Device isolation: The Receipt Service hides printer-specific commands and the Payment Service hides the card-terminal API. Changing hardware does not require rewriting touchscreen interaction or order validation.
2. Manageable scope: Three coffee types, two sizes and one payment method need a straightforward order workflow. Separating UI, coordination and adapters makes this small system easier to develop, test and maintain.

Security Advantage

The Order Manager obtains authoritative prices from the Menu Repository instead of trusting UI totals. Only the Payment Service accesses the terminal/gateway; other components receive a transaction reference and approval status, never raw card details. Restricting credentials to this boundary reduces exposure. These controls must be implemented; layering alone does not guarantee security.

Performance Benefit

Menu reads and internal component calls remain local, avoiding network round trips between kiosk modules. A small validated menu cache can keep browsing responsive. Payment runs asynchronously so the touchscreen remains responsive while approval is pending; payment latency still depends on the external system.

Order and Failure Handling

The Order Manager validates coffee and size, calculates the price, and sends the order ID and total for payment. A declined payment produces no success receipt. A timeout is reconciled using the same transaction identifier before retrying to avoid duplicate charges. After approval, a printer failure offers a reprint of the paid order without charging again.